PROJECT ASSIGNMENT

IDS 2023-2024 , M1 MOSIG & M1 INFO

Here are the detailed explanations on what is expected of you for your project and how it will be graded.

General considerations

What you are supposed to work on is the design and implementation of a *distributed* system. This means that the core of your work is the identification and the definition of the entities that will be part of your system (distributed entities) and of their (distributed) interactions.

You should be very clear about the architecture of the system (which entity does what and how does it interact with the others, under what conditions). Once this step is over, it will be very easy to find a suitable (working) technical solution, be it in RMI or RabbitMQ.

What you should turn in

Every group should turn in (on Moodle) a tar.gz file named **`NAME1_NAME2.tar.gz`** with:

- [compulsory] a report, containing
- [compulsory] the GIT URL of your project
- [optional] compilation/launching scripts
- [optional] libraries, jar files, packages, etc.

Even if the last points are optional, you **must provide an explanation of how to run the system**.

The report

The report is the master piece.

GENERAL ADVICE ABOUT THE REPORT

- The report should be written in correct French/English (use the dedicated tools for that)
- The report should explicitly give your names
- Number the pages
- Explain correctly without thinking that "we know". We prefer that you explain as if we do not know.
- We want pictures, schemes for the design, interactions.

The report should be organized as follows :

- Architecture
  - Do not get mixed between system architecture and code structure
  - What we want to see are the running entities, their roles, the interactions
  - Describe your assumptions (i.e. your project targets some situations, but not others)
  - Justify your choices based on simplicity, logic, realism, performance, ...
  - Describe the limitations you are aware of or you have chosen (because of time, cost, performance, ...)
  - If it is in RMI, we want to see what is Remote and what are the interfaces
  - If it is in RabbitMQ, we want to see the queues (with explanation of their specificities)
- Implementation
  - Here is where you should present the software structure of your project (e.g which class implements what), as well as details about what you really did
  - If there is a GUI, we need to be able to find the code corresponding to the core functionality and it is much better that this code be separated from the interface management code
- Deployment
  - Please do explain how we can compile, install and run your system.
  - Do not forget the distribution i.e. the basic model would likely be "*one entity = one virtual machine*" (be it Python or Java). If you use threads to "make it simpler" you have the risk of forgetting that you are to be distributed and in a distributed setting there is no shared memory (no commun variables, files, etc).
  - You can provide scripts fro launching the system or the different entities.
  - Beware! We do not have the same environment as you. You need to include configurations or explanations of how to do them concerning the paths, the libraries, the dependencies, etc.
  - The **`mandlebrot`** server is considered to be a reference environment. If your project runs on mandelbrot, this is perfect (and in this case do specify this in your report).

    You can access mandelbrot though ssh
    `ssh <your_login>@mandelbrot.e.ujf-grenoble.fr`

    If you need to redirect a graphical interface you should use
    `ssh -X <your_login>@mandelbrot.e.ujf-grenoble.fr`

- Conclusion
  - We would like you to step back and comment on your project.
  - The comments may concern the degree of maturity of your project, the difficulties you had, some reflexions on the technology, ideas for new projects, performance considerations, ...
  - and finally, if you think that you learned something useful and whether you are glad with what you did :)

The code

Be sure to get rid of all temporary files, as well as all IDE-related files.

For Java, if you want to turn in compiled files, please turn in jars and not class files.

Defenses

15 minutes is really really short and it goes by without you noticing it!

Explain the major points of the project, show us that it works (screen sharing) and answer some questions directly.

You do not have the time to explain everything about your project during the defense session !

You should have several slides presenting a general overview of the architecture, the entities and the interactions (~5 min), show us how it runs (~5 min) and then we will discuss (~5 min, i.e. 2-3 questions).

Grading

This is an initial version of the grading rules.

The general trend is

- 1/3 of the points for a functional project (is it doing what it is supposed to be doing, is it doing it well?),
- 1/3 for a good presentation of the project (report + live session)
- and 1/3 on the software engineering aspects (the choices you made, you implemented).

Which gives something like :

Aspect	Points
Project	6
Report	6
Sound architecture	2
Quality code	2
Deployment rules/tools	2
Overall maturity	2
TOTAL	20

The "maturity" points are related to you being able to reason about your project like being able to comment on some choices being simplistic, some possible improvements, etc.